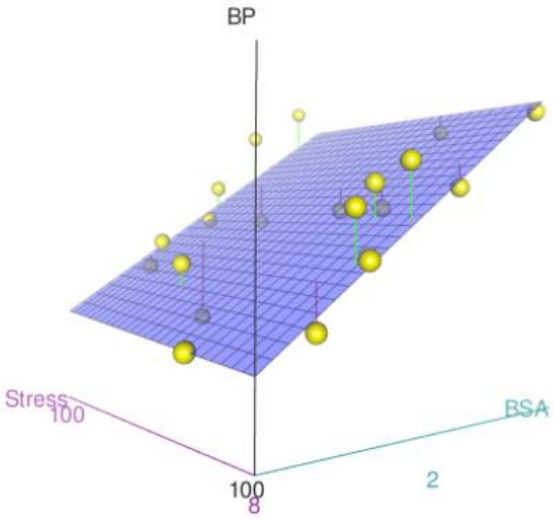
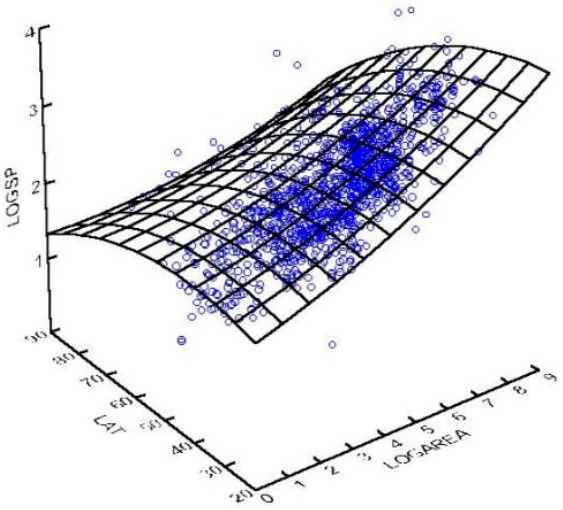
Chapter 6.1: Multiple Linear Regression (MLR)

Multiple Linear Regression **vs** **Nonlinear multiple linear regression**





Ordinary Least Squares (OLS)

The Ordinary Least Squares (OLS) approach to multiple linear regression was introduced by Gauss in 1794. The OLS procedure is the simplest type of estimation procedure used in statistical analyses. However, in order to benefit from the well-behaved properties of an OLS estimate, a number of assumptions must be satisfied.

In matrix notation, the model is written as:

$$y = X\beta + \varepsilon \qquad (2)$$

Assumptions

The estimate of the regression parameters in Equation (2) is denoted as $\hat{\beta}$. There are five assumptions necessary to produce unbiased estimators using OLS. Additional assumptions must be satisfied in order for the estimate to have other favorable qualities.

The assumptions for OLS estimation are:

1. The model must be linear in the parameters.

2. The data are a random sample of the population i.e. residuals are statistically independent/uncorrelated from each other
   i.e. $\text{Cov}(\varepsilon_i, \varepsilon_i') = 0 \; \forall \; \text{i} \neq \text{i'}$.

3. The independent variables are not too strongly collinear.
4. The independent variables are measured precisely such that measurement error is negligible.

5. The expected value of the residuals is always zero
   i.e. $\mathbb{E}(\varepsilon) = 0$.

6. The residuals have constant variance (homogeneous variance)
   i.e. $\mathbb{V}(\varepsilon) = \sigma^2 \mathbb{I}$.
   (Tests for this include the Bruesch-Pagan test and Brown-Forsythe test).

7. The residuals are Normally distributed i.e. $\varepsilon \sim N(0, \sigma^2 \mathbb{I})$.

Chapter 6.4

Use Adjusted R-Squared to Include the Correct Number of Variables

What Is the Adjusted R-squared?

- The adjusted R-squared compares the explanatory power of regression models that contain different numbers of predictors.
- It can be helpful in model selection.
- Suppose you compare a five-predictor model with a higher R-squared to a one-predictor model.
- Does the five predictor model have a higher R-squared because it's better? Or is the R-squared higher because it has more predictors? Simply compare the adjusted R-squared values to find out!
- The adjusted R-squared is a modified version of R-squared that has been adjusted for the number of predictors in the model.

$$R^2 = 1 - \frac{SS_{residuals}}{SS_{total}}$$

$$\text{Adjusted } R^2 = 1 - \frac{SS_{residuals} / (n-K)}{SS_{total} / (n-1)}$$

K is the number of parameters fit by the regression.

- The adjusted R^2 always has a lower value than R^2 (unless you are fitting only one parameter). The equations above show why.
- The adjusted *R*-squared **increases** only if the new term improves the model more than would be expected by chance.

- It **decreases** when a predictor improves the model by less than expected by chance. The adjusted *R*-squared can be negative, but it's usually not. It is always lower than the *R*-squared.
- In the simplified Best Subsets Regression output below, you can see where the adjusted *R*-squared peaks, and then declines. Meanwhile, the *R*-squared continues to increase.

Vars	R-Sq	R-Sq(adj)
1	72.1	71.0
2	85.9	84.8
3	87.4	85.9
4	89.1	82.3
5	89.9	80.7

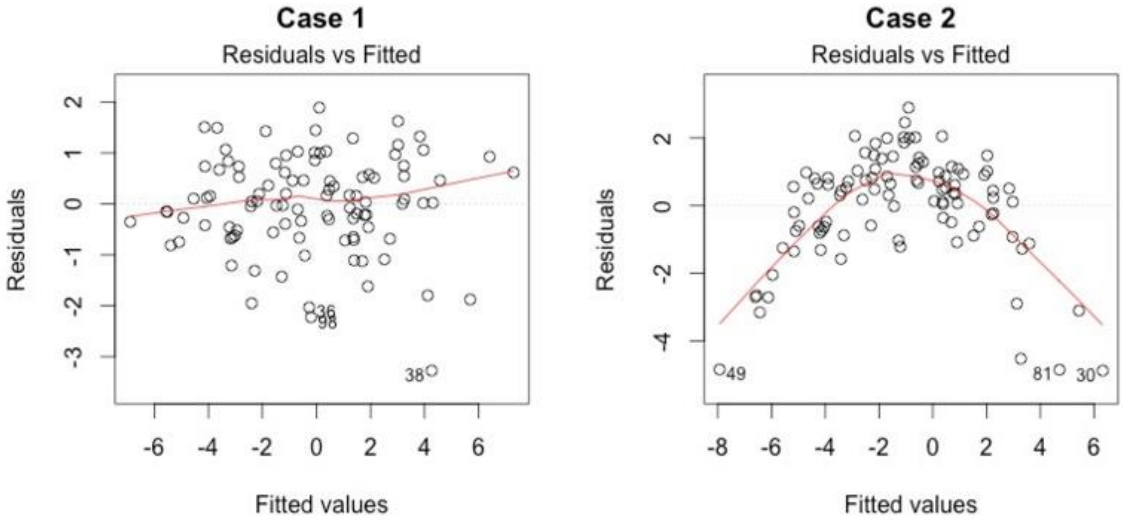
- You might want to include only three predictors in this model.

Understand Diagnostic Plots for Linear regression Analysis

1. Residuals vs Fitted

This plot shows if residuals have non-linear patterns. There could be a non-linear relationship between predictor variables and an outcome variable and the pattern could show up in this plot if the model doesn't capture the non-linear relationship. If you find equally spread residuals around a horizontal line without distinct patterns, that is a good indication you don't have non-linear relationships.

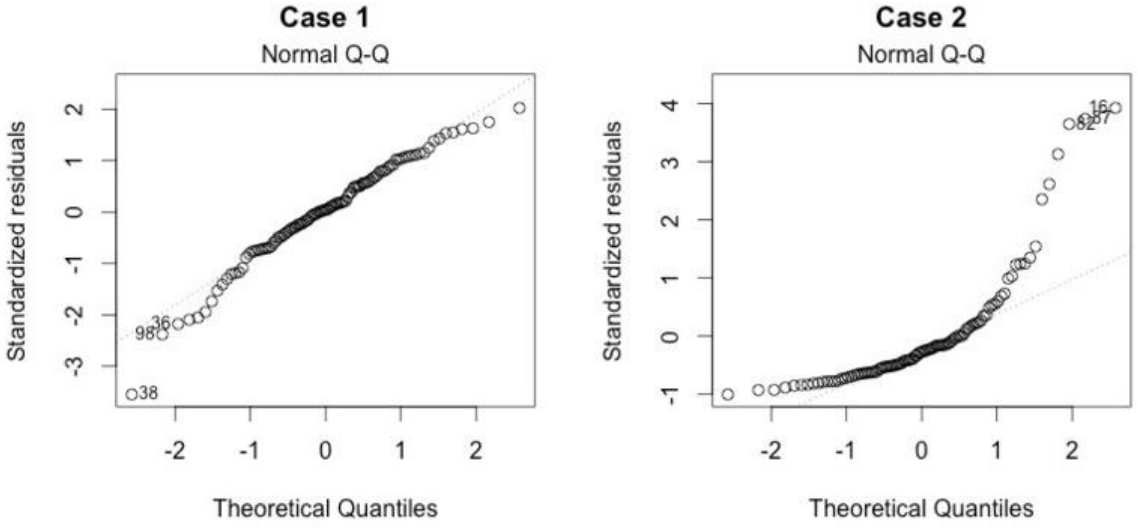
Let's look at residual plots from a 'good' model and a 'bad' model. The good model data are simulated in a way that meets the regression assumptions very well, while the bad model data are not.



No distinctive pattern in Case 1, but a parabola in Case 2, where the non-linear relationship was not explained by the model and was left out in the residuals.

2. Normal Q-Q

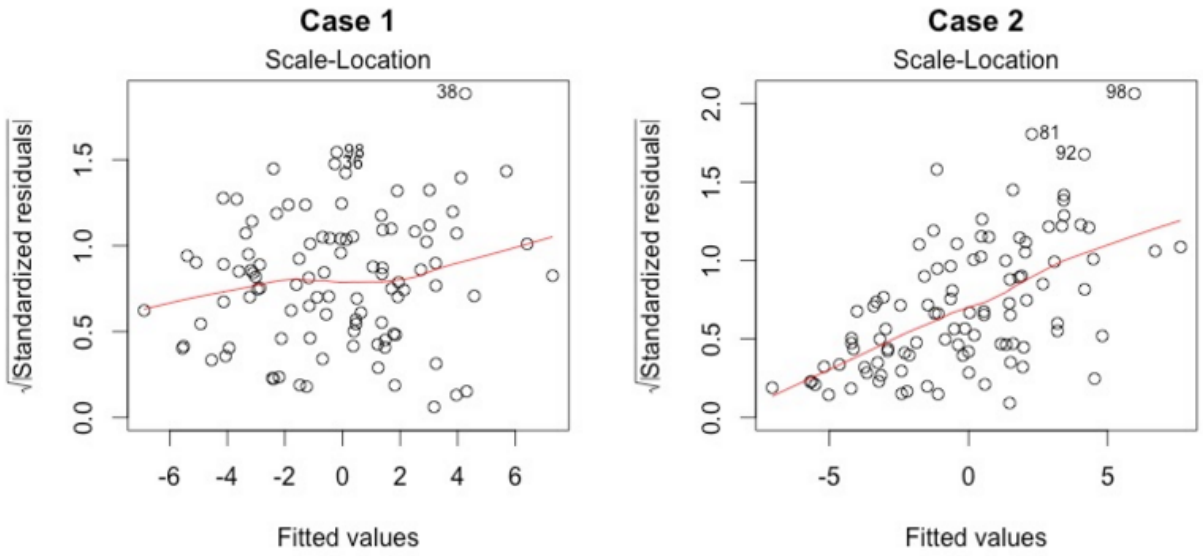
This plot shows if residuals are normally distributed. Do residuals follow a straight line well or do they deviate severely? It's good if residuals are lined well on the straight dashed line.



Of course they wouldn't be a perfect straight line. Case 2 definitely concerns us , we would not be concerned by Case 1 too much, although an observation numbered as 38 looks a little off. Let's look at the next plot while keeping in mind that #38 might be a potential problem.

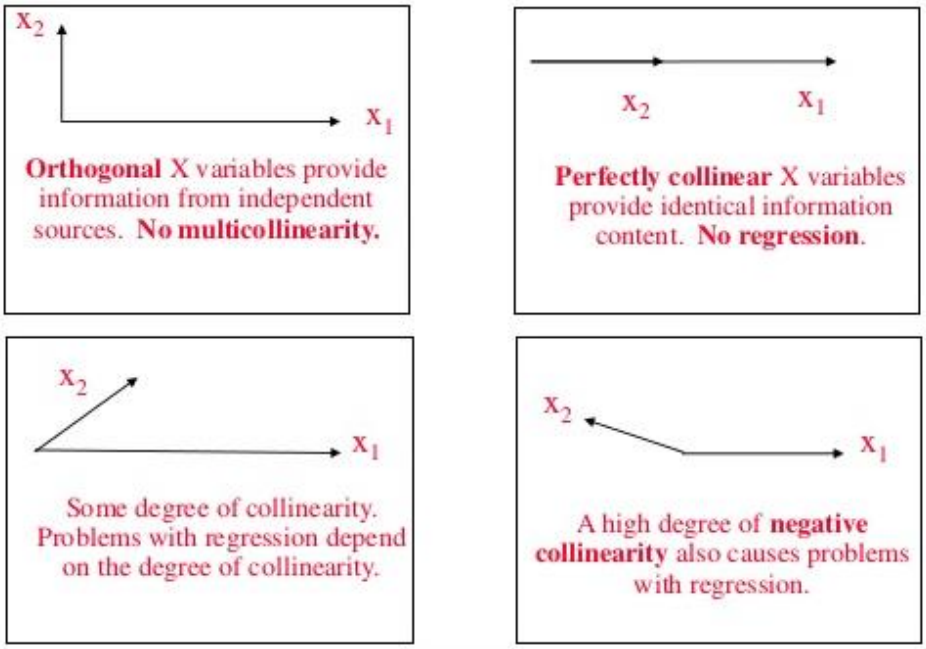
3. Scale-Location

It's also called Spread-Location plot. This plot shows if residuals are spread equally along the ranges of predictors. This is how you can check the assumption of equal variance (homoscedasticity). It's good if you see a horizontal line with equally (randomly) spread points.



In Case 1, the residuals appear randomly spread. Whereas, in Case 2, the residuals begin to spread wider along the x-axis as it passes around 5. Because the residuals spread wider and wider, the red smooth line is not horizontal and shows a steep angle in Case 2.

Multicollinearity



What is "Multicollinearity"?

- Multicollinearity (or inter correlation) exists when at least some of the predictor variables are correlate among themselves.
- A "linear" relation between the predictors. Predictors are usually related to some extent; it is a matter of degree.
- Multicollinearity cannot be tested; only the degrees of multicollinearity can be detected.
- Multicollinearity is a feature of the sample and not of the population. Therefore, we do not "test for multicollinearity" but we measure its degree in any particular sample.
- The **Variance Inflation Factor (VIF)** measures the impact of collinearity among the variables in a regression model. There is no formal VIF value for determining presence of multicollinearity. Values of VIF that exceed 10 are often regarded as indicating multicollinearity, but in weaker models values above 2.5 may be a cause for concern.

$$VIF\left(\hat{\beta}_k\right) = \frac{1}{1 - R_k^2}$$

Chapter 6.5

Given that

$$\hat{y_i} = \hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \hat{\beta}_2 x_{i,2} + \ldots + \hat{\beta}_k x_{i,k}$$

Interpretation of the Model Parameters

- Each β coefficient represents the change in the mean response, E(y), per unit increase in the associated predictor variable when all the other predictors are held constant.
- For example, β_1 represents the change in the mean response, E(y), per unit increase in x_1 when x_2, x_3, ..., x_k are held constant.
- The intercept term, β_0, represents the mean response, E(y), when all the predictors x_1, x_2, ..., x_k, are all zero (which may or may not have any practical meaning).

Example:

$$Y = 42 + 2.3X_1 + 11I(X_2)$$

The above multiple linear regression model predicts the height of a specific plant (Y) in cm based on:

- X_1: The amount of bacteria in the soil (measured in 1000 per ml of soil).
- X_2: An indicator variable for sunlight condition, where:
  - $I(X_2) = 0$ if the plant is in **partial sunlight**
  - $I(X_2) = 1$ if the plant is in **full sunlight**

➢ **Continuous variable X_1 (Bacteria)**
Holding the sunlight condition constant, an increase of **1000 bacteria per ml of soil** is associated with an expected **increase of 2.3 cm** in plant height, on average.

➢ **Categorical variable X_2 (Sunlight condition)**
A plant in **full sunlight** ($I(X_2) = 1$) is expected to be **11 cm taller on average than a plant in partial sunlight** ($I(X_2) = 0$), assuming the bacteria level remains constant.

➢ **Intercept** ($\beta_0 = 42$)
The intercept of 42 cm is the predicted height when $X_1 = 0$ (no bacteria) and $I(X_2) = 0$ (partial sunlight). As noted, this interpretation is meaningful only if the values $X_1 = 0$ and $I(X_2) = 0$ are realistic for the data. If bacteria levels never approach 0, the

intercept simply serves as a mathematical anchor for the model rather than having a practical interpretation.

In this case, since X_2 (sunlight condition) can take the value 0, that part of the interpretation remains valid. However, if X_1 (bacteria level) never comes close to 0 in the dataset, then the intercept does not have a meaningful interpretation in practical terms—it simply ensures the correct positioning of the regression equation.

Remark

- **Correlated explanatory variables**

  Each coefficient in a regression model represents the effect of its corresponding variable **while holding other variables constant**. However, in real-world data, explanatory variables are often correlated, meaning that they may share some explanatory power over *Y*.

  As a result, **each coefficient does not measure the total effect of that variable on *Y*** but rather its unique contribution after accounting for the influence of other predictors in the model.

How to write a fitted model with both continuous and categorical explanatory variables?

1.

Fitted model : (No interaction)

$$\hat{y} = \begin{cases} \hat{\beta}_0 + \hat{\beta}_1 x & \text{if type} = 1 \\ (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 x & \text{if type} = 2 \end{cases}$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 \, I\{\text{type} = 2\}$$

Indicator function

type 1 : 'baseline', or 'reference' level

type 2 : additional effect

In R: lm(y ~ x + type)

2.

Fitted model : (With interaction effect)

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 \, I\{\text{type} = 2\} + \hat{\beta}_3 x \, I\{\text{type} = 2\}$$

$$= \begin{cases} \hat{\beta}_0 + \hat{\beta}_1 x & \text{if type} = 1 \\ (\hat{\beta}_0 + \hat{\beta}_2) + (\hat{\beta}_1 + \hat{\beta}_3) x & \text{if type} = 2 \end{cases}$$

In R: lm(y ~ x * type)